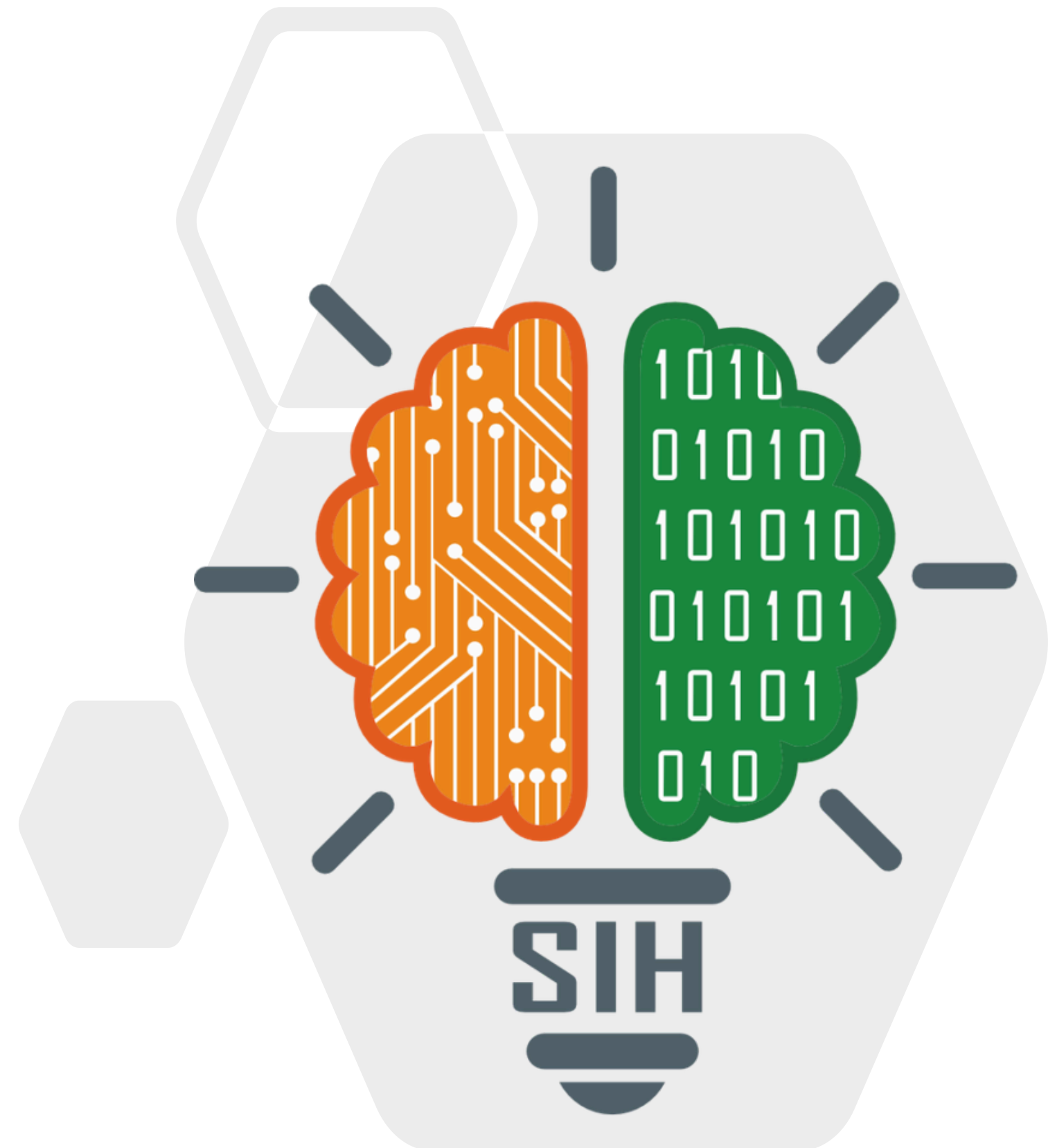


SMART INDIA HACKATHON 2026

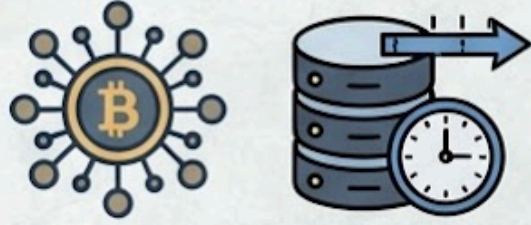


- **Problem Statement ID –26146**
- **Problem Statement Title- AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic**
- **Theme-Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID-**
- **Team LATENT**



PROPOSED SOLUTION

Proposed Solution: Dual-Layer Bitcoin Transaction Forensics



Secure Nobody Closes: Ledger records what moved, not who. **STRATA** joins ledger and network data on time.

The Gap Nobody Closes: Ledger records what moved, not who. **STRATA** joins ledger and network data on time.



Explainable Scoring: Every lead carries verifiable evidence, decomposed by SHAP.

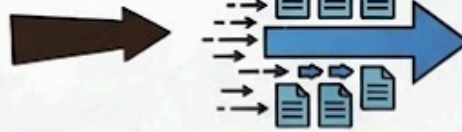
Transparent Oversight: Admin Dashboard provides real-time activity monitoring.



Addressing Key Problems



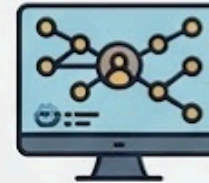
Bulk Ingest & GeoIP: Parses metadata at ~40,000 records/sec entirely offline.



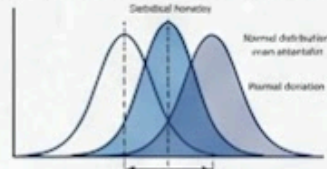
Entity Resolution: Recovers laundering peeling chains and filters CoinJoins.



Attribution: First-relay counts tested against host baseline.

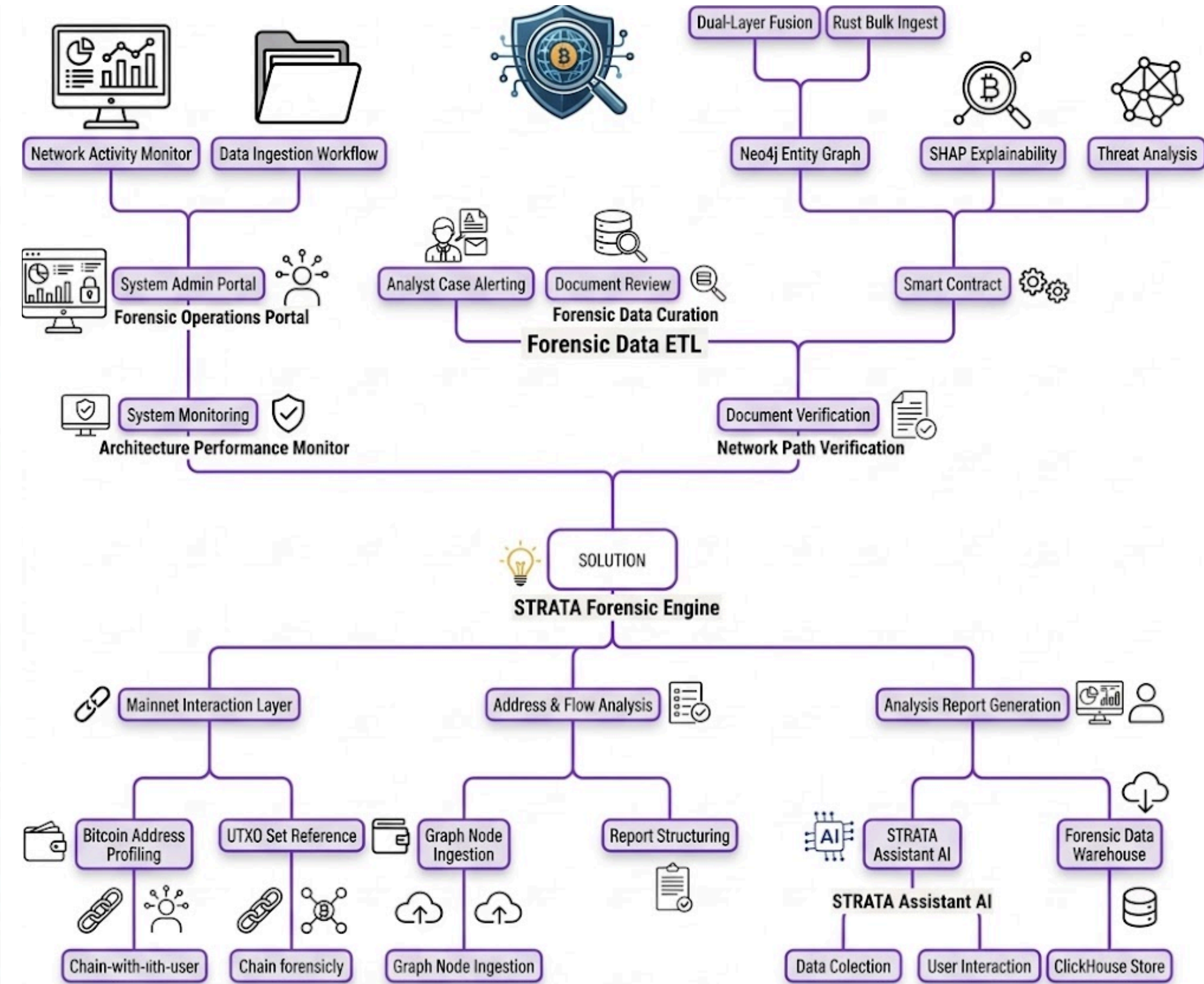


Innovation & Uniqueness



- **Statistical honesty:** Exact binomial tail, corrected for all pairs.

- **Repetition is the evidence:** One sighting proves nothing.
- **Statistical honesty:** Exact binomial tail, corrected for all pairs.
- **Negative controls:** Shuffle the labels and the system finds nothing.
- **Fully Offline:** No outbound requests. Keeping queries secret.



A Crucial Security Check: STRATA identifies illicit flows only after the SHAP analysis confirms the feature vector matches the known illicit pattern on the **Bitcoin Mainnet**, protecting the analysis from adversarial obfuscation.



Forensics Case File

TECHNICAL APPROACH

Team
Latent



1 • **Generator** synthesises the dataset and ground truth, so detection can be measured rather than asserted.

2 • **Ingest (Rust):** CSV/JSON/XML into one shape at ~40,000 records/sec. integer satoshis, never floats. SHA-256 custody manifest.

Data Source & Ground Truth

(Step 1)

Step 2: Transaction

Step 2: Wallet State Ingest

Ingest Wallet data and Satoshis

CoinJoins excluded to prevent false merges

Step 3: Clustering
- CoinJoin filter & CIOH

Recovers laundering structures

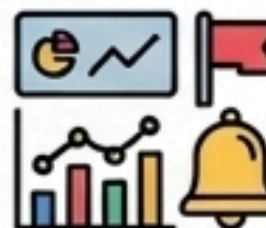
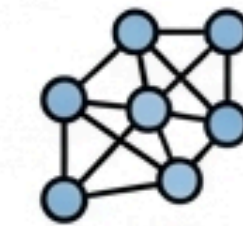
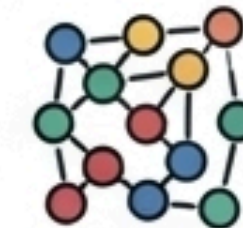
Step 4: Graph
Peeling chains recovered

Dual-layer correlation on time

Step 5: Fusion
First-relay vs baseline

Scores decomposed by SHAP

Step 6: Detection
RF, IsolationForest & SHAP



DEPLOYED LINK:

<https://strata-alpha-two.vercel.app/>

Ingest   **Parquet**
csv, serde, json, quick-xml, rayon, maxminddb, arrow/parquet.
Chosen for bulk throughput and memory safety.

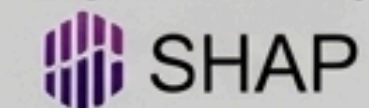
Analysis



Detection

scikit-learn
RandomForest +
IsolationForest.

Explainability



Exact on trees, not
an approximation.

Geolocation



DB-IP Lite country +
ASN .mmdb, shipped
in-repo.

Dashboard

Streamlit-aggrid -
st-link-analysis -
Altair.

Landing page

Three.js
(vendored, no CDN).

Store  **DuckDB**
Apache Parquet + DuckDB.
Columnar, embedded, zero
server.

Analysis



5 • Fusion

first-relay counts vs the host's
baseline, exact binomial;
Bonferroni-corrected.

6 • Detection

RandomForest +
IsolationForest, every score
decomposed by SHAP.

FEASIBILITY AND VIABILITY



Feasibility Analysis

- **Ready for Review** – Already built and measured six stages, end to end, with a live deployment a judge can open
- **Proven Accuracy:** – 66 of 67 automated checks pass, each scored against the generator's ground truth



Viability

- **Resource Efficient:** – Commodity hardware, no GPU required. Full rebuild completes in seconds, not hours
- **Highly Accessible:** – No licence cost relies entirely on an open-source stack and a GeoIP database under CC-BY



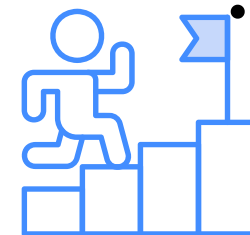
Challenges and Risks

- **CoinJoin Poisoning:** – Strangers co-signing transactions risk merging innocent parties into false suspect clusters
- **Multiple Comparisons:** – Testing hundreds of entity pairs statistically generates false leads by design
- **Air-Gapped Constraints:** Strict requirement for zero cloud APIs, zero lookups, and zero outbound requests



Overcoming Challenges

- **Precision Filtering** – CoinJoins excluded early; 100% precision on evaluation
- **Statistical Correction** – Bonferroni correction dropped false leads to 0
- **Calculated Mergesy:**– Stage declines risky guesses; accuracy boosted to 98%.
- **Offline Construction:**– Zero outbound requests; everything ships in-repo



Supporting Facts

Law Enforcement Triage: Replaces messy spreadsheets with ranked, explainable queues

FIU & Traceability: Exposes multi-hop laundering and peeling chains cleanly.



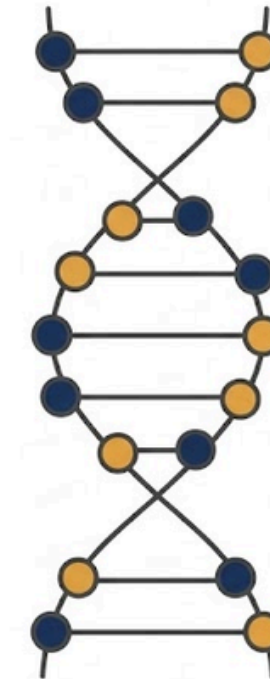
Accuracy Evaluation

Measured against ground truth from synthetic data.



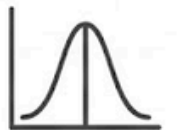
Deployment Viability

Commodity hardware, no GPU, zero outbound requests.



CoinJoin Exclusion

100% precision on evaluation set, zero false merges.



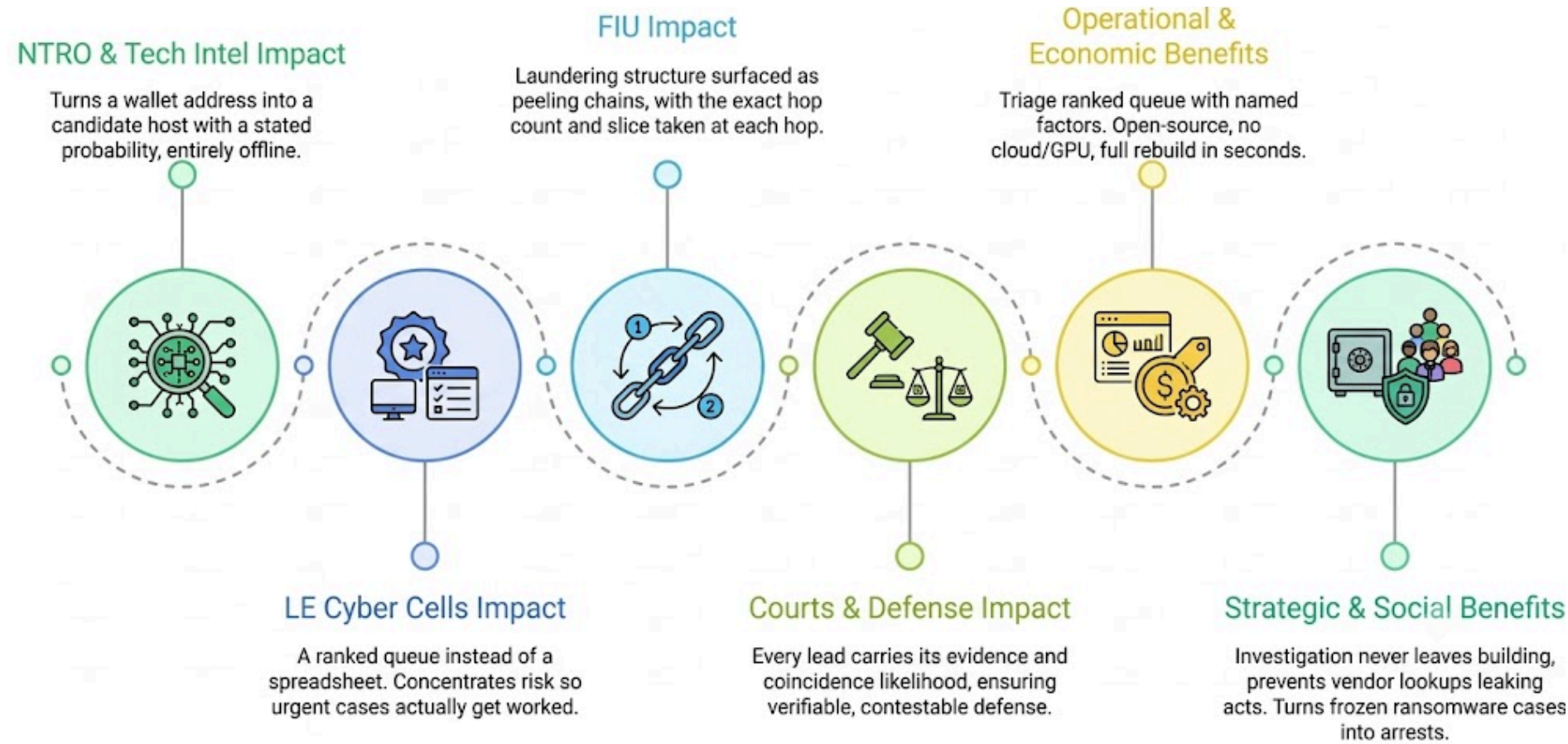
Statistical Correction

Bonferroni correction reduces false leads from 41 to 0.



Business Potential and Use Cases

- **Law Enforcement** –Replaces raw logs with a ranked, explainable triage queue.
- **Financial Units** – Surfaces laundering structures and peeling hop counts.
- **Regulatory Defense** – Delivers verifiable evidence and statistical confidence scores
- **National Security**-Complete air-gapped protection preventing vendor data leaks



Benefits of the solution:



✳ Operational Benefits -

- Analysts triage a ranked queue instead of reading raw transaction logs
- Every score arrives with named factors; analysts open cases knowing why



✳ Economic Benefits -

- Open-source, no per-seat licence, no cloud spend, no GPU required
- Runs on commodity hardware; a full rebuild takes seconds

✳ Strategic / National Security -

- The investigation never leaves the building.
- Prevents foreign vendor GeoIP lookups from leaking the act of investigating



✳ Technological Benefits -

- Air-Gapped & Sovereign:** Zero outbound requests and zero cloud APIs; the forensic pipeline runs entirely offline inside your secure environment.
- High-Throughput Safety:** Leverages memory-safe Rust for bulk ingestion (~40k records/sec) combined with embedded DuckDB for zero-server columnar analysis.



Potential impacts on target audience:

- NTRO & Tech Intel**– Turns a wallet address into a candidate host with a stated probability, entirely offline.
- Law-Enforcement Cyber Cells** –A ranked queue instead of a spreadsheet. Concentrates risk so urgent cases actually get worked.
- Financial Intelligence Units** – Laundering structure surfaced as peeling chains, with the exact hop count and slice taken at each hop.
- Courts & The Accused** –Every lead carries its evidence and coincidence likelihood, ensuring verifiable, contestable defense
- Trust & Accountability** – Immutable records and verification build transparency for all.

Project Links & Demonstration -

Live Deployment

- <https://strata-alpha-two.vercel.app/>

Demo Video

- To be added

Source Code

- <https://github.com/aryanraj45/strata.git>

Further Reading & Key Resources -

Network-layer deanonymisation

- Biryukov, Khovratovich & Pustogarov Deanonymisation of Clients in Bitcoin P2P Network, ACM CCS 2014. The first-relay estimator our fusion stage builds on.
- Koshy, Koshy & McDaniel An Analysis of Anonymity in Bitcoin Using P2P Network Traffic, Financial Cryptography 2014

Ledger-layer entity resolution

- Androulaki, Karame, Roeschlin, Scherer & Capkun - Evaluating User Privacy in Bitcoin, Financial Cryptography 2013.
- Ron & Shamir - Quantitative Analysis of the Full Bitcoin Transaction Graph, Financial Cryptography 2013.

Data and standards

- DB-IP Lite country & ASN databases (CC-BY 4.0) shipped offline in-repo.
- RFC 5737 — IPv4 address blocks reserved for documentation, used for every synthetic network address.